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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE

EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCFSQ100 - FOOD AND NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skil(S), Interest(I) andAppreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. c) 1 [K]/[CO1, CO2]
2. b) Synthesized by the body [U]/[CO1]
3. b) It provides real-time data to refine dietary recommendations[U]/[CO1, CO4]
4. b) Fortified plant-based milk [U]/[CO1]
5. b) Undernutrition [K]/[CO1]
6. b) Build and repair tissues [K]/[CO1]
7. d) Vitamin B9 [U]/[CO1, CO3]
8. d) All of the above [K]/[CO1]
9. a) Vitamin K [K]/[CO1]
10. c) Vitamin D [K]/[CO1, CO3, CO4]

Part B

Short essay

Answer any **4** questions

Each question carries **5** marks

11. Give the definitions for nutrition and health. [K]/[CO1]

Nutrition:

- Nutrition is the science of food, nutrients, and their effect on the body.
- It includes ingestion, digestion, absorption, utilization, and excretion of food.

- Proper nutrition is essential for growth, energy, repair, and maintenance of health.

Health:

- Health is a state of complete physical, mental, and social well-being.
- It is not merely the absence of disease.
- Good health enables an active and productive life

12. Summarize the symptoms of iodine deficiency disorder(IDD).[U]/[CO1,CO3,CO4]

Iodine deficiency affects thyroid hormone production and causes several disorders. Main symptoms include:

- **Goitre** – enlargement of thyroid gland causing neck swelling.
- **Hypothyroidism** – tiredness, weight gain, dry skin, slow metabolism.
- **Weakness and fatigue**.
- **Poor concentration and low mental alertness**.
- **Growth retardation in children**.
- **Delayed puberty**.
- **Mental impairment in children**.
- During pregnancy, severe deficiency may cause **miscarriage, stillbirth, or cretinism** in babies

13. List the main food groups included in a plant-based diet. [K]/[CO1, CO4]

A plant-based diet mainly includes foods obtained from plant sources. Main food groups are:

- **Whole grains** – rice, wheat, oats, maize, millet.
- **Pulses and legumes** – beans, peas, lentils, chickpeas, soybeans.
- **Vegetables** – leafy vegetables, roots, tubers, other vegetables.
- **Fruits** – banana, apple, orange, mango, berries etc.
- **Nuts and seeds** – almonds, walnuts, sesame, flaxseed, sunflower seeds.
- **Plant oils** – olive oil, coconut oil, groundnut oil.
- **Herbs and spices** for flavor and health benefits.

14. Explain on the methods of measurement of BMR. [U]/[CO1, CO2]

Basal Metabolic Rate (BMR) is the minimum energy required by the body at complete rest. It can be measured by the following methods:

1. Direct Calorimetry

- Measures heat produced by the body.
- Person stays inside a calorimeter chamber.
- Heat released indicates metabolic rate.
- Accurate but costly and complex.

2. Indirect Calorimetry

- Measures oxygen consumption and carbon dioxide production.
- Used to estimate energy expenditure.
- Commonly used in hospitals and laboratories.

3. Predictive Equations

- BMR estimated using formulas such as Harris-Benedict equation.
- Based on age, sex, height, and body weight.
- Useful in diet planning.

15. Write short notes on general symptoms of carbohydrate deficiency. [U]/[CO1]

Carbohydrates are the main source of energy. Deficiency may produce the following symptoms:

- Weakness and tiredness.
- Fatigue during work or exercise.
- Dizziness and headache.
- Poor concentration and irritability.
- Weight loss due to breakdown of body tissues.
- Ketosis due to excess fat breakdown.
- Muscle wasting if proteins are used for energy.
- Constipation if fiber intake is also low.
- Reduced physical and mental performance.

16. Discuss the role of macronutrients (proteins, carbohydrates and fats) in space nutrition. [U]/[CO1, CO4]

Macronutrients are very important for astronauts during space travel.

Proteins

- Maintain muscles and body tissues.
- Prevent muscle loss in microgravity.
- Needed for enzymes, hormones, and immunity.
- Help repair damaged tissues.

Carbohydrates

- Main source of energy.
- Supply glucose for brain and muscles.
- Improve work performance and alertness.
- Spare proteins from being used as energy.

Fats

- Concentrated source of energy.
- Provide essential fatty acids.
- Help absorb vitamins A, D, E, and K.
- Support cell membrane and hormone function.

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

17. Elaborate on Polysaccharides. [K]/[CO1]

Polysaccharides are complex carbohydrates composed of many monosaccharide units joined together by glycosidic bonds. They usually contain more than ten sugar units and may consist of hundreds or even thousands of glucose molecules. Polysaccharides are widely distributed in plants, animals, and microorganisms. They are important in nutrition because they serve as energy reserves, structural materials, and dietary fiber. Compared to simple sugars, polysaccharides are less sweet, less soluble, and digested more slowly.

Classification of Polysaccharides

Polysaccharides are mainly classified into **homopolysaccharides** and **heteropolysaccharides**.

1. Homopolysaccharides

These contain only one type of monosaccharide unit repeated many times. Important examples are:

- **Starch:** Main storage carbohydrate of plants. Found in rice, wheat, maize, potato, and tapioca. It is the major energy source in the human diet.
- **Glycogen:** Storage carbohydrate of animals. Stored mainly in liver and muscles. It acts as a reserve source of glucose.
- **Cellulose:** Structural carbohydrate present in plant cell walls. It is indigestible to humans and acts as dietary fiber.

2. Heteropolysaccharides

These contain two or more different types of monosaccharides. Examples include:

- **Pectin:** Present in fruits and used in jam and jelly making.
- **Hemicellulose:** Present in plant cell walls and contributes to dietary fiber.
- **Gums and mucilages:** Used as thickening and stabilizing agents in foods.

Important Components of Starch

Starch consists of two fractions:

- **Amylose:** Linear chain of glucose molecules. Responsible for gel formation.
- **Amylopectin:** Highly branched chain. Responsible for viscosity and thickening properties.

Nutritional Importance

- Major source of energy.
- Digested slowly and provides sustained energy.
- Helps maintain blood glucose levels.
- Dietary fiber components improve bowel movement.
- Increase satiety and reduce overeating.

Functional Importance in Food Industry

- Used as thickener in soups, sauces, gravies.
- Gives texture and body to foods.
- Used in bakery and confectionery products.
- Pectin forms gels in jams and jellies.
- Gums stabilize ice cream and beverages.

Thus, polysaccharides are essential carbohydrates having nutritional, physiological, and industrial importance. They serve as energy stores, structural materials, and functional ingredients in many foods.

18. Explain the role of Vitamin A in maintaining overall health of a person. [K]/[CO1]

Vitamin A is a fat-soluble vitamin essential for growth, vision, immunity, reproduction, and maintenance of epithelial tissues. It occurs as retinol in animal foods and as beta-carotene in plant foods. Vitamin A is required in small amounts but plays a major role in maintaining overall health.

Sources of Vitamin A

Animal sources:

- Liver

- Egg yolk
- Butter
- Milk
- Fish liver oils

Plant sources (carotene):

- Carrot
- Mango
- Papaya
- Pumpkin
- Spinach
- Green leafy vegetables

Functions of Vitamin A

1. Vision

Vitamin A forms rhodopsin pigment in the retina, which is essential for vision in dim light. It prevents night blindness.

2. Maintains Healthy Skin and Epithelial Tissue

It keeps skin, respiratory tract, digestive tract, and urinary tract linings healthy and resistant to infection.

3. Supports Immunity

Vitamin A enhances resistance against infections and improves immune response.

4. Growth and Development

Essential for normal growth of bones and body tissues in children.

5. Reproduction

Needed for normal reproductive health in both males and females.

6. Antioxidant Role

Beta-carotene acts as an antioxidant and protects cells from oxidative damage.

Deficiency Symptoms

- Night blindness
- Xerophthalmia
- Dry skin
- Poor growth
- Frequent infections
- Bitot's spots

Vitamin A is essential for maintaining vision, immunity, growth, healthy skin, and overall well-being. Adequate intake through diet is necessary to prevent deficiency disorders.

19. Explain the concept of climate friendly nutrition and its importance in reducing environmental impact. [U]/[CO1, CO4]

Climate-friendly nutrition refers to dietary choices and food systems that promote human health while reducing negative environmental effects such as greenhouse gas emissions, excessive water use, land degradation, and food waste. It focuses on selecting foods that are nutritious, sustainable, and environmentally responsible.

Modern food production contributes significantly to climate change through methane emission, deforestation, pollution, and waste generation. Climate-friendly nutrition aims to create diets that protect both human health and the planet.

Main Principles of Climate-Friendly Nutrition

1. Preference for Plant-Based Foods

Diets rich in cereals, pulses, fruits, vegetables, nuts, and seeds generally have lower environmental impact than meat-based diets.

2. Reduced Intake of Resource-Intensive Foods

Lower consumption of red meat and heavily processed foods helps reduce greenhouse gas emissions.

3. Use of Local and Seasonal Foods

Local foods require less transportation and refrigeration, reducing fuel use and emissions. Seasonal foods are fresher and economical.

4. Reduction of Food Waste

Avoiding food waste saves water, land, energy, and labor used in food production.

5. Sustainable Farming Practices

Supports organic farming, crop rotation, efficient irrigation, and reduced chemical use.

Importance in Reducing Environmental Impact

- Reduces greenhouse gas emissions.
- Conserves land, water, and energy resources.
- Protects biodiversity and ecosystems.
- Reduces pollution caused by packaging and transport.
- Improves long-term food security.
- Encourages responsible consumption habits.

Health Benefits

Climate-friendly diets are usually rich in fiber, vitamins, minerals, and antioxidants. They reduce risk of obesity, diabetes, and heart disease.

Climate-friendly nutrition is an important approach that links health with sustainability. It helps individuals eat healthier foods while protecting the environment for future generations.

20. Explain in detail about the sources, distribution and functions of water.

[U]/[CO1, CO2]

Water is an essential nutrient required for life. It forms a major part of the human body and is necessary for all physiological activities. Though it provides no calories, water is indispensable for survival, metabolism, and maintenance of body functions.

Sources of Water

Water is obtained from three main sources:

1. Drinking Water

The major source of body water comes from plain water and beverages such as milk, juice, tea, soups, and other drinks.

2. Water Present in Foods

Many foods contain water naturally. Fruits and vegetables have high water content. Examples: watermelon, cucumber, orange, tomato.

3. Metabolic Water

Small amount of water is produced inside the body during oxidation of carbohydrates, fats, and proteins.

Distribution of Water in the Body

Water constitutes about 50–70% of body weight depending on age, sex, and body composition.

Intracellular Fluid (ICF)

- About two-thirds of total body water.
- Present inside body cells.

Extracellular Fluid (ECF)

- About one-third of total body water.
- Includes blood plasma, lymph, interstitial fluid, digestive juices, and cerebrospinal fluid.

Functions of Water

1. Solvent and Medium for Reactions

Water acts as a universal solvent and medium for metabolic reactions.

2. Transport of Nutrients

Carries nutrients, oxygen, hormones, and waste materials through blood and body fluids.

3. Regulation of Body Temperature

Maintains body temperature through sweating and evaporation.

4. Digestion and Absorption

Helps in digestion, absorption, and utilization of nutrients.

5. Excretion of Waste Products

Removes waste through urine, sweat, and feces.

6. Lubrication

Lubricates joints, eyes, and tissues.

7. Maintains Cell Shape and Blood Volume

Supports cellular structure and maintains blood pressure and circulation.

Water Balance

Water lost through urine, sweat, feces, and respiration must be replaced daily. Dehydration results when water loss exceeds intake.

Water is a vital nutrient necessary for transport, digestion, temperature control, waste removal, and life processes. Adequate daily intake is essential for maintaining health and normal body function.